Formulas & Tricks — Made Easy

Units 3 & 4

VCE Mathematical Methods Units 3 & 4 · Functions · Trigonometry · Transformations · Probability · Normal distribution · Sample proportions

This is the short, friendly version of the big formula sheet. Every rule comes with a picture, a table or a worked example. Look for the coloured boxes:

TRICK — saves time | TRAP — where marks are lost | CHECK — catch your own mistakes | EXAMPLE — see it worked out

1 Functions and graphs

Interval notation — square bracket or round bracket?

Write	Means	On a graph	Example
`[a, b]`	both ends included	● filled dots at both ends	`[1, 4]` → $1 \le x \le 4$
`(a, b)`	both ends left out	○ open circles at both ends	`(1, 4)` → $1 < x < 4$
`[a, b)`	only a included	● at a, ○ at b	`[1, 4)` → $1 \le x < 4$
`R \ {2}`	every real number except 2	hole / asymptote at x = 2	`(−∞, 2) ∪ (2, ∞)`

∞ always gets a round bracket — you can never "reach" infinity.

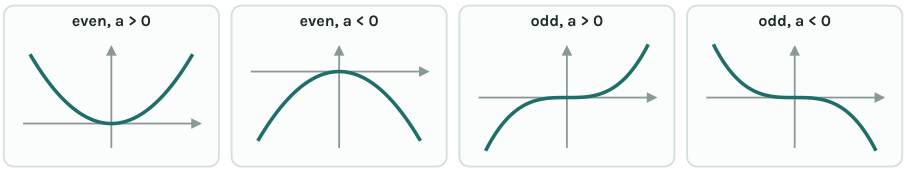
Maximal (largest possible) domain

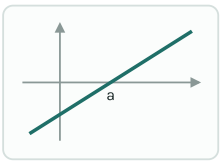
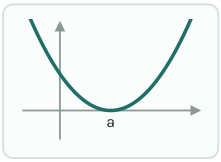
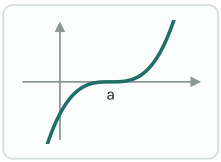
If the rule has…	You need…	Example	Domain
a square root √(g(x))	`g(x) ≥ 0`	`√(x − 3)`	`[3, ∞)`
a fraction 1 / g(x)	`g(x) ≠ 0`	`1 / (x + 5)`	`R \ {−5}`
a log $\log_e(g(x))$	`g(x) > 0`	$\log_e(2 - x)$	`(−∞, 2)`
more than one of these	all of them at once	see the example below	

EXAMPLE `y = √(x + 2) + 1/(x − 1)`. Root: x ≥ −2. Fraction: x ≠ 1. Together: **`[−2, 1) ∪ (1, ∞)`**.

Polynomials — shape from the degree

A polynomial of degree **n** has **at most n** x-intercepts and **at most n − 1** turning points. Only the **leading term** decides where the arms go:



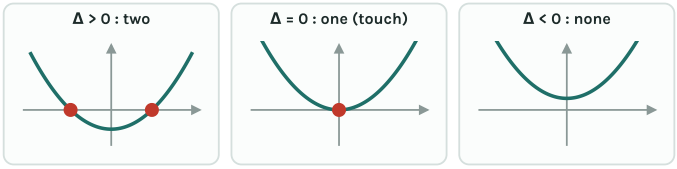
Factor	Picture near x = a	What the graph does	Sign changes?
$(x - a)$		**cuts** straight through	yes
$(x - a)^2$		**touches** and turns back (turning point)	no
$(x - a)^3$		**flattens** then goes through (stationary point of inflection)	yes

Rule of thumb: **even** power → touch, **odd** power → cross.

EXAMPLE $y = (x - 1)^2(x + 3)$: touches the x-axis at x = 1, cuts at x = −3. It is a cubic but has only **2 distinct** zeros.

Quadratics $y = ax^2 + bx + c$

What you want	Formula	Quick note
x of the turning point	$x = -b / (2a)$	then substitute to get y
turning point form	$y = a(x - h)^2 + k \rightarrow TP\ (h, k)$	watch the minus: $y = 2(x + 3)^2 - 1$ has TP (−3, −1)
quadratic formula	$x = (-b \pm \sqrt{(b^2 - 4ac)}) / (2a)$	leave surds exact
axis of symmetry	$x = (p + q) / 2$	halfway between the x-intercepts p and q
discriminant	$\Delta = b^2 - 4ac$	tells you **how many** solutions — see pictures



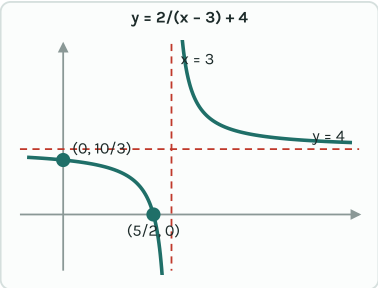
TRICK

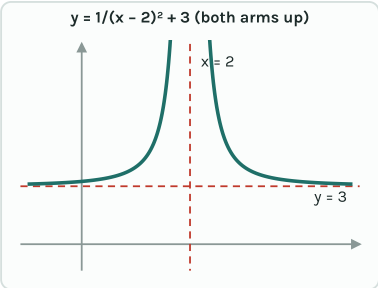
Use the discriminant — do not solve

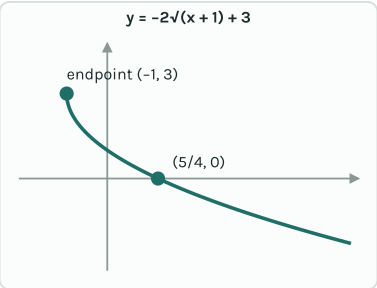
"For what k is the line a tangent?" → set the rules equal, collect into $ax^2 + bx + c = 0$, then put **$\Delta = 0$**. ($\Delta > 0$: two intersections, $\Delta < 0$: none.) No need to actually solve the quadratic.

Hyperbola, truncus and square-root graphs

Graph	Rule	Asymptotes / endpoint	Domain	Range
Hyperbola	$y = a/(x - h) + k$	x = h, y = k	$R \setminus \{h\}$	$R \setminus \{k\}$
Truncus	$y = a/(x - h)^2 + k$	x = h, y = k	$R \setminus \{h\}$	(k, ∞) if $a > 0$
Square root	$y = a\sqrt{(x - h)} + k$	endpoint (h, k)	$[h, \infty)$	$[k, \infty)$ if $a > 0$ $(-\infty, k]$ if $a < 0$







EXAMPLE $y = 2/(x - 3) + 4$ x-intercept: $2/(x - 3) = -4 \to x - 3 = -\frac{1}{2} \to x = 5/2$. y-intercept: $2/(-3) + 4 = 10/3$.

TRICK

Split the numerator instead of long division

Make the top contain the bottom: $(3x + 1)/(x - 2) = (3(x - 2) + 7)/(x - 2) = 3 + 7/(x - 2)$. Now you can read the asymptotes straight off: **x = 2** and **y = 3**.

CHECK

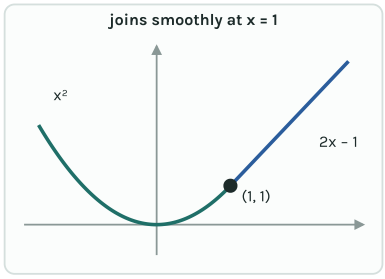
Endpoint or asymptote? Pick the right bracket

A square-root endpoint **is** reached → square bracket: range of $y = -2\sqrt{(x + 1)} + 3$ is $(-\infty, 3]$. An asymptote is **never** reached → round bracket / "\": range of the truncus above is $(3, \infty)$.

Hybrid (piecewise) functions

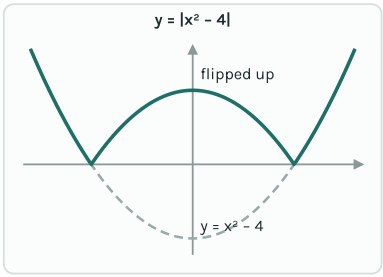
To be…	Check at the join x = a
continuous (no gap)	left value = right value = f(a)
differentiable (smooth, no corner)	continuous **and** left gradient = right gradient

EXAMPLE $f(x) = x^2$ for $x \le 1$, $2x - 1$ for $x > 1$.
Values: $1^2 = 1$ and $2(1) - 1 = 1$ ✓ Gradients: $2x = 2$ and 2 ✓
→ smooth join.
If it were x^2 and x: values match ($1 = 1$) but gradients $2 \neq 1$ → a **corner**, not differentiable.



Modulus |x| and odd / even functions

Type	Rule	Example
equation	$\lvert x - a\rvert = b \to x = a \pm b$	$\lvert x - 2\rvert = 5 \to x = 7$ or -3
"less than"	$\lvert x\rvert < a \Leftrightarrow -a < x < a$	$\lvert x\rvert < 3 \to (-3, 3)$
"greater than"	$\lvert x\rvert > a \Leftrightarrow x < -a$ or $x > a$	two separate pieces
graph of $\lvert f(x)\rvert$	flip everything **below** the x-axis up	
graph of $f(\lvert x\rvert)$	keep $x \ge 0$, mirror it in the y-axis	



Even function	Odd function
$f(-x) = f(x)$ — mirror image in the y-axis. e.g. x^2, cos x, $\lvert x\rvert$	$f(-x) = -f(x)$ — same after a half-turn about the origin. e.g. x^3, sin x, $1/x$

TRAP

Double factor touches, triple factor flattens

At $(x - a)^2$ the curve turns back. At $(x - a)^3$ it flattens and **keeps going**. Students often draw a turning point at a triple factor — don't.

CHECK

Count features against the degree

A degree-n polynomial can't have more than n x-intercepts or more than n − 1 turning points. A quartic drawn with 4 turning points is wrong on sight.

Try it yourself — Section 1

1. Find the maximal domain of $y = \sqrt{(4 - x)} + 1/(x + 1)$.
2. Find the asymptotes of $y = (2x + 5)/(x + 1)$. (Hint: split the numerator.)
3. Describe what $y = (x + 2)^2(x - 1)^3$ does at $x = -2$ and at $x = 1$.

2 Trigonometric functions

Radians and exact values

`π rad = 180°` · degrees → radians: × π/180 · radians → degrees: × 180/π

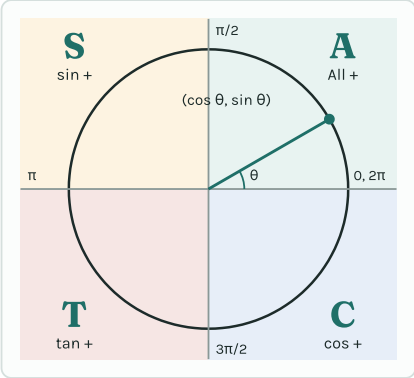
θ	0	π/6 (30°)	π/4 (45°)	π/3 (60°)	π/2 (90°)
sin θ	0	1/2	√2/2	√3/2	1
cos θ	1	√3/2	√2/2	1/2	0
tan θ	0	√3/3 (= 1/√3)	1	√3	undefined

TRICK

Remember the sin row in 5 seconds

Write `√0/2, √1/2, √2/2, √3/2, √4/2` → 0, ½, √2/2, √3/2, 1. The cos row is the **same numbers backwards**. tan = sin ÷ cos.

The unit circle and the quadrants (ASTC)



"All Students Take Calculus" — which ratio is positive

Quadrant	Angle in terms of base angle θ	Positive
1 (0 to π/2)	`θ`	all
2 (π/2 to π)	`π − θ`	sin only
3 (π to 3π/2)	`π + θ`	tan only
4 (3π/2 to 2π)	`2π − θ`	cos only

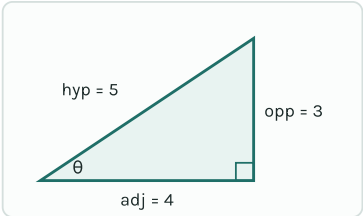
EXAMPLE sin(7π/6): 7π/6 = π + π/6 → quadrant 3, base angle π/6. sin is negative there, so **sin(7π/6) = −1/2**.

Also useful: `sin(−θ) = −sin θ`, `cos(−θ) = cos θ`, `sin(π/2 − θ) = cos θ`, `sin²θ + cos²θ = 1`, `tan θ = sin θ / cos θ`.

Finding one ratio from another

EXAMPLE sin θ = 3/5 and θ ∈ (π/2, π). Find cos θ and tan θ.

① Draw the triangle: opp 3, hyp 5 → adj 4 (3-4-5).
② Quadrant 2: cos is negative → **cos θ = −4/5**.
③ **tan θ = (3/5) ÷ (−4/5) = −3/4**.



the triangle gives the size, the quadrant gives the sign

TRAP

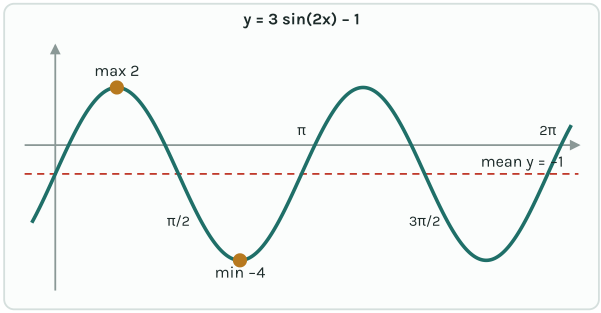
The identity gives the size, the quadrant gives the sign

cos²θ = 1 − 9/25 = 16/25 gives cos θ = ±4/5. Only the given domain tells you which sign. Writing +4/5 loses the mark even though the arithmetic is right.

Graphs of sine and cosine: $y = a\sin(n(x + b)) + c$

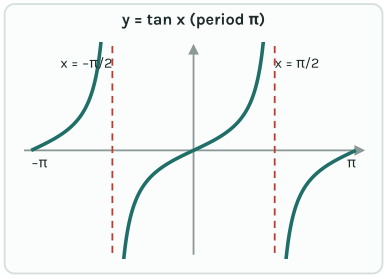
Feature	Formula	For $y = 3\sin(2x) - 1$
amplitude	$\lvert a\rvert$	3
period	$2\pi / \lvert n\rvert$	$2\pi/2 = \pi$
mean (centre) line	$y = c$	$y = -1$
range	$[c - \lvert a\rvert, c + \lvert a\rvert]$	$[-4, 2]$
cycles in $[0, 2\pi]$	$2\pi \div$ period	2 cycles

Key points are a **quarter-period apart**: sine starts at the mean line going up; cosine starts at a max (if $a > 0$).



The tangent graph $y = a\tan(n(x + b)) + c$

Feature	Rule
period	$\pi / \lvert n\rvert$
asymptotes of tan x	$x = \pi/2 + k\pi,\ k \in Z$
x-intercepts of tan x	$x = k\pi,\ k \in Z$
amplitude	**none** — range is R



EXAMPLE $y = \tan(2x)$: period $\pi/2$. Asymptotes: $2x = \pi/2 + k\pi \rightarrow x = \pi/4 + k\pi/2$.

Solving a trig equation on a restricted domain — 4 steps

Example: solve $2\sin(2x) + 1 = 0$ for $x \in [0, 2\pi]$.

1 Isolate the trig ratio
$2\sin(2x) + 1 = 0 \rightarrow \sin(2x) = -\frac{1}{2}$

→

2 Change the domain
$x \in [0, 2\pi] \rightarrow 2x \in [0, 4\pi]$

→

3 Base angle → quadrants → add periods
base $\pi/6$; sin − in Q3, Q4: $7\pi/6$, $11\pi/6$, then $+2\pi$: $19\pi/6$, $23\pi/6$

→

4 Undo (÷ 2)
$x = 7\pi/12, 11\pi/12, 19\pi/12, 23\pi/12$

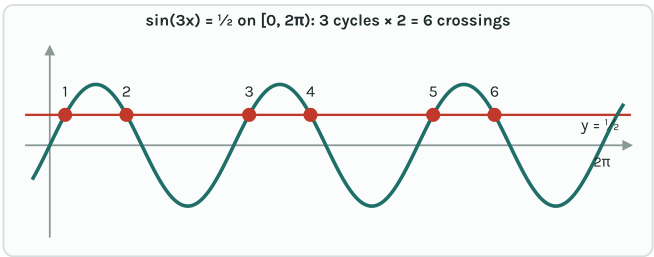
TRICK

Change the domain, not the answers

Convert the domain first ($2x \in [0, 4\pi]$), find **every** answer for $2x$ in that bigger interval, then divide by 2 at the very end. Solving for x first and adding periods later is where solutions go missing.

How many solutions should I expect?

Equation (over one interval of length 2π)	Number of solutions	Example on $[0, 2\pi)$
$\sin(nx) = k$ or $\cos(nx) = k$, with $\lvert k\rvert < 1$	**2n**	$\sin(3x) = \frac{1}{2} \rightarrow$ 6 solutions
$\sin(nx) = \pm 1$ or $\cos(nx) = \pm 1$	**n** (one per cycle)	$\sin(2x) = 1 \rightarrow$ 2 solutions
$\tan(nx) = k$ (any k)	**n per π**, so 2n per 2π	$\tan x = -\sqrt{3} \rightarrow$ 2 solutions
sin or cos = k with $\lvert k\rvert > 1$	**0**	$\sin x = 2 \rightarrow$ no solution



EXAMPLE sin(3x) = ½ on [0, 2π): 3x ∈ [0, 6π). Base angle π/6, sin + in Q1, Q2:
3x = π/6, 5π/6, 13π/6, 17π/6, 25π/6, 29π/6 → **x = π/18, 5π/18, 13π/18, 17π/18, 25π/18, 29π/18** — six answers, matching the graph. **If you have fewer, you dropped some.**

CHECK

Range check kills impossible equations

y = 3 sin(2x) − 1 only takes values in [−4, 2]. So `3 sin(2x) − 1 = 3` has **no solution** — check this before you start solving.

General solutions (n ∈ Z)

Equation	General solution	Example
`sin x = a`	`x = 2nπ + sin⁻¹(a) or x = (2n + 1)π − sin⁻¹(a)`	`sin x = ½ → x = 2nπ + π/6 or (2n+1)π − π/6`
`cos x = a`	`x = 2nπ ± cos⁻¹(a)`	`cos x = ½ → x = 2nπ ± π/3`
`tan x = a`	`x = nπ + tan⁻¹(a)`	`tan x = 1 → x = nπ + π/4`

CHECK

Put n = 0, 1, −1 back in

Always write **n ∈ Z**. Then substitute n = 0, 1, −1 — the values that land in the domain should match your restricted-domain answers.

TRAP

Factorise before you read the translation

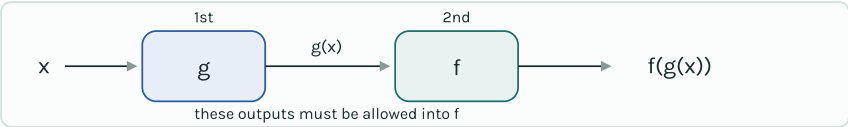
`sin(2x − π/3)` is **not** a shift of π/3. Write it as `sin(2(x − π/6))`: the shift is **π/6 to the right**. Any time n ≠ 1, take n out of the bracket first.

Try it yourself — Section 2

1. How many solutions does `cos(2x) = 0.3` have for x ∈ [0, 2π]?
2. Solve `cos x = −½` for x ∈ [0, 2π].
3. State the amplitude, period and range of `y = −2 cos(3x) + 5`.

3 Composite functions and transformations

Composite functions f(g(x)) — think of two machines



Fact	Rule	What it means
Order	`f(g(x))`	do **g first**, then f (also written f ∘ g)
Exists when	`ran g ⊆ dom f`	everything g puts out must be allowed into f
Domain	`dom f(g(x)) = dom g`	after any restriction you had to make
Not symmetric	`f(g(x)) ≠ g(f(x))`	one can exist while the other does not

EXAMPLE — IT FAILS, THEN FIX IT $f(x) = \sqrt{x}$, $g(x) = x - 3$ (dom R).
ran g = R, dom f = [0, ∞). R ⊄ [0, ∞) → f(g(x)) **does not exist**.
Fix: restrict dom g to [3, ∞). Then ran g = [0, ∞) ✓ and $f(g(x)) = \sqrt{(x - 3)}$.

EXAMPLE — RANGE OF A COMPOSITE $f(x) = x^2$, $g(x) = x - 1$, dom g = [0, 4].
Step 1: ran g = [−1, 3].
Step 2: square that set → **[0, 9]**, **not** [1, 9], because x = 1 gives 0, the minimum.

TRAP

It is ran g ⊆ dom f, not dom g ⊆ dom f

The test is on the **outputs** of the inside function. Write down ran g, write down dom f, then compare those two sets.

The transformation rule y = a f(b(x + c)) + d

Letter	Effect	Exam wording	Example from y = f(x)
a	multiplies every **y** by a (a < 0 also flips)	dilation of factor |a| from the x-axis	`y = 3f(x): 3 × taller`
b	**divides** every **x** by b (b < 0 also flips)	dilation of factor 1/|b| from the y-axis	`y = f(2x): squashed, factor ½`
c	moves left by c	translation of c units in the negative x direction	`y = f(x + 3): 3 left`
d	moves up by d	translation of d units in the positive y direction	`y = f(x) + 2: 2 up`

Image of a point: `(p, q) → (p/b − c, aq + d)`. **Order:** dilations and reflections first, translations last.

Example: y = 2f(3(x − 1)) + 5 sends the point (2, 4) to …

Coordinate	Working	Result
x	divide by 3, then add 1 → 2/3 + 1	**5/3**
y	multiply by 2, then add 5 → 2 × 4 + 5	**13**
Check with $f(x) = x^2$: $2(3(5/3 - 1))^2 + 5 = 2(2)^2 + 5$		= 13 ✓

TRAP

Dilation factors from the y-axis go upside down

f(2x) is a dilation of factor ½ from the y-axis, not 2 — the graph is squashed towards the y-axis. Vertical ones read straight off: 2f(x) is factor 2 from the x-axis.

TRICK

Asymptotes and endpoints move too

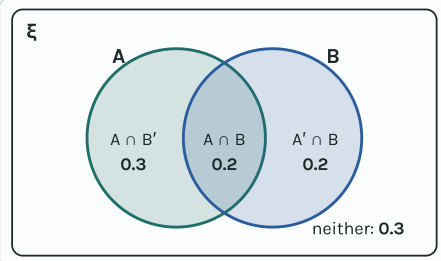
Apply the point rule to x = h and y = k as well. For y = 1/x → y = 2/(x − 3) + 4, the asymptotes x = 0, y = 0 move to **x = 3, y = 4**. Fastest way to sketch any transformed graph.

Try it yourself — Section 3

1. f(x) = √x and g(x) = x + 2 with dom g = R. Does f(g(x)) exist? If not, give the largest domain of g that fixes it.
2. Find the image of the point (1, 3) under `y = −f(2(x + 1)) + 4`.
3. Describe the transformations from y = sin x to `y = sin(2x − π/3)`.

4 Probability rules and conditional probability

Rule	Formula
Complement	`Pr(A′) = 1 − Pr(A)`
Addition ("or")	`Pr(A ∪ B) = Pr(A) + Pr(B) − Pr(A ∩ B)`
Mutually exclusive	`Pr(A ∩ B) = 0`
Conditional ("given")	`Pr(A | B) = Pr(A ∩ B) / Pr(B)`
Multiplication ("and")	`Pr(A ∩ B) = Pr(A | B) × Pr(B)`
Independent	`Pr(A ∩ B) = Pr(A) × Pr(B)`
Neither	`Pr(A′ ∩ B′) = 1 − Pr(A ∪ B)`



Pr(A) = 0.5, Pr(B) = 0.4, Pr(A ∩ B) = 0.2

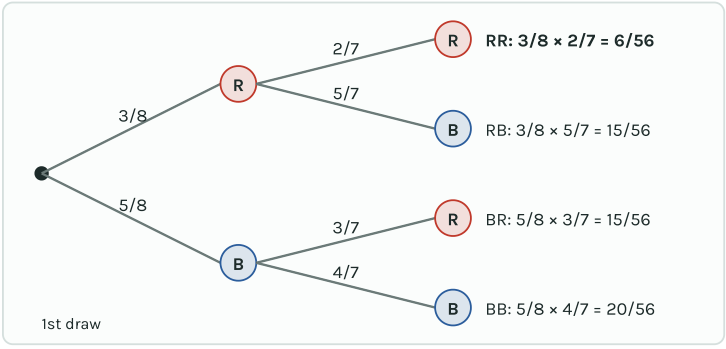
The two-way (Karnaugh) table — fastest method for "and / or / given / neither"

	B	B′	Total
A	**0.2**	0.5 − 0.2 = **0.3**	0.5
A′	0.4 − 0.2 = **0.2**	**0.3**	0.5
Total	0.4	0.6	**1**

EXAMPLE Pr(A ∪ B) = 0.5 + 0.4 − 0.2 = **0.7**. Neither = **0.3**. Pr(A | B) = 0.2 ÷ 0.4 = **0.5** = Pr(A) → A and B are **independent**. (Check: 0.5 × 0.4 = 0.2 ✓)

Fill the totals you are given, then the inner cells. Every row and column must add up, and the corner is always 1.

Tree diagrams — multiply along, add across



3 red and 5 blue counters, draw 2 **without** replacement

	With replacement	Without replacement
Pr(both red)	`3/8 × 3/8 = 9/64`	`3/8 × 2/7 = 3/28`
Probabilities	stay the same	change each draw
Model	binomial ✓	tree, **not** binomial

Pr(one of each) = RB + BR = 15/56 + 15/56 = 30/56 = 15/28. Branches leaving any point must add to 1.

TRAP

Mutually exclusive is NOT the same as independent

Mutually exclusive: A and B **can't both happen** (Pr(A ∩ B) = 0). Independent: one **doesn't change** the chance of the other. If both events can happen on their own, mutually exclusive events are always **dependent**. Always prove independence with numbers: compare Pr(A ∩ B) with Pr(A) × Pr(B).

Try it yourself — Section 4

1. Pr(A) = 0.6, Pr(B) = 0.5 and Pr(A ∩ B) = 0.3. Find Pr(A ∪ B) and Pr(A′ ∩ B′).
2. Are A and B in Q1 independent? Show why.

5 Discrete random variables and the binomial distribution

LINK TO YEAR 7 — COLLECTING & SUMMARISING DATA

E(X) is the "mean from a frequency table" you already know

In Year 7: mean = Σ(score × frequency) ÷ total. Divide each frequency by the total and you get a probability — so the mean becomes $E(X) = \Sigma\, x \cdot Pr(X = x)$.

Books read, x	0	1	2	3	Total
Frequency	6	6	4	1	17
Pr(X = x)	6/17	6/17	4/17	1/17	1

Year 7: (0×6 + 1×6 + 2×4 + 3×1) ÷ 17 = 17 ÷ 17 = 1. Methods: 0(6/17) + 1(6/17) + 2(4/17) + 3(1/17) = **1**. Same answer!

A probability distribution — the one-table method

Two rules: every probability is between 0 and 1, and they **add to 1**. Then build E(X) and E(X²) in the same table:

x	0	1	2	3	Sum
Pr(X = x)	0.1	0.3	**k = 0.4**	0.2	1 → k = 0.4
x · Pr	0	0.3	0.8	0.6	**E(X) = 1.7**
x² · Pr	0	0.3	1.6	1.8	**E(X²) = 3.7**

$Var(X) = E(X^2) - [E(X)]^2 = 3.7 - 1.7^2 = 0.81 \rightarrow sd(X) = \sqrt{0.81} = 0.9$

TRAP

E(X²) is not [E(X)]²

Square each **x**, keep the probabilities the same. If your variance comes out negative, you squared the wrong thing.

Changing X: aX + b

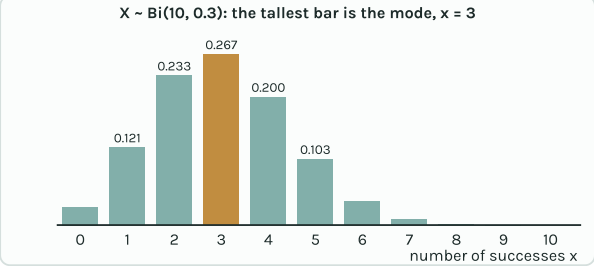
Measure	Rule	E(X) = 2.1, Var(X) = 0.49 → for 3X − 1
Mean	$E(aX + b) = aE(X) + b$	3(2.1) − 1 = **5.3**
Variance	$Var(aX + b) = a^2\, Var(X)$	9 × 0.49 = **4.41** (the −1 does nothing)
Standard deviation	$sd(aX + b) = \lvert a \rvert\, sd(X)$	3 × 0.7 = **2.1**

The binomial distribution X ~ Bi(n, p)

Use it only when **all four** are true:

n	**2**	**p**	**⊥**
fixed number of trials	outcomes: success / failure	same chance every trial	trials independent

What	Formula
Probability	$Pr(X = x) = {}^nC_x\, p^x\, (1 - p)^{n-x}$
Mean	$E(X) = np$
Variance	$Var(X) = np(1 - p)$
At least one	$Pr(X \geq 1) = 1 - (1 - p)^n$
Most likely value	mode = whole-number part of $(n + 1)p$



(n + 1)p = 11 × 0.3 = 3.3 → mode 3. Mean np = 3, variance 2.1

EXAMPLE $X \sim Bi(5, 0.2)$: $Pr(X = 2) = {}^5C_2(0.2)^2(0.8)^3 = 10 \times 0.04 \times 0.512 =$ **0.2048**.
Check: the powers 2 + 3 add to n = 5 ✓

Turning words into inequalities

The question says…	Write	Values included (x = 0 to 6)	Calculator form
exactly 3	`Pr(X = 3)`	0 1 2 **3** 4 5 6	`binomPdf(n, p, 3)`
at most 3 / no more than 3	`Pr(X ≤ 3)`	**0 1 2 3** 4 5 6	`binomCdf(n, p, 0, 3)`
fewer than 3 / less than 3	`Pr(X ≤ 2)`	**0 1 2** 3 4 5 6	`binomCdf(n, p, 0, 2)`
at least 3 / no fewer than 3	`Pr(X ≥ 3) = 1 − Pr(X ≤ 2)`	0 1 2 **3 4 5 6**	`binomCdf(n, p, 3, n)`
more than 3	`Pr(X ≥ 4)`	0 1 2 3 **4 5 6**	`binomCdf(n, p, 4, n)`
between 2 and 5 inclusive	`Pr(2 ≤ X ≤ 5)`	0 1 **2 3 4 5** 6	`binomCdf(n, p, 2, 5)`

TRAP
Off by one in "at least"
For a discrete X: `Pr(X ≥ 3) = 1 − Pr(X ≤ 2)`, **not** 1 − Pr(X ≤ 3). For Bi(20, 0.4) the right answer is 0.9964; the wrong one gives 0.9840 — close enough to look fine, still wrong.

TRICK
"At least one" is one line
Never add Pr(X = 1) + Pr(X = 2) + … Use `1 − (1 − p)^n`. With n = 5, p = 0.2: $1 - 0.8^5$ = **0.67232**.
Backwards — "how many trials for at least a 95% chance?": $1 - 0.8^n \geq 0.95 \rightarrow n \geq 13.4 \rightarrow$ **n = 14** (always round **up**; n = 13 only gives 0.945).

TRICK
Conditional binomial: divide, don't subtract
`Pr(X ≥ 2 | X ≥ 1) = Pr(X ≥ 2) ÷ Pr(X ≥ 1)` because "X ≥ 2" sits inside "X ≥ 1". For Bi(8, 0.25): 0.6329 ÷ 0.8999 = **0.7033**.

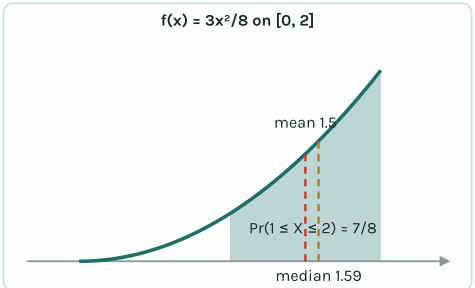
CHECK
Binomial variance is always smaller than the mean
np(1 − p) < np. Bi(20, 0.4): mean 8, variance 4.8. If you get a variance of 8 or more, you forgot the (1 − p).

Try it yourself — Section 5

1. x = 1, 2, 3 with Pr = 0.2, k, 0.5. Find k and E(X).
2. $X \sim Bi(10, 0.1)$. Find E(X), Var(X) and $Pr(X \geq 1)$.
3. Write "more than 2 but fewer than 6" as an inequality in X.

6 Continuous random variables (probability density functions)

What	Formula
It is a pdf if	`f(x) ≥ 0 and total area = 1`
Probability = area	`Pr(a ≤ X ≤ b) = ∫ₐᵇ f(x) dx`
Mean	`E(X) = ∫ x f(x) dx`
Variance	`E(X²) − [E(X)]², E(X²) = ∫ x² f(x) dx`
Median m	`∫ from start to m of f(x) dx = 0.5`
Mode	x where f(x) is highest (can be an endpoint)



For a continuous X, Pr(X = a) = 0, so ≤ and < give the same answer.

EXAMPLE f(x) = 3x²/8 on [0, 2].
Area: [x³/8] from 0 to 2 = 8/8 = 1 ✓ · Pr(1 ≤ X ≤ 2) = 8/8 − 1/8 = **7/8**
E(X) = ∫ 3x³/8 dx = [3x⁴/32] = 48/32 = **3/2** · E(X²) = [3x⁵/40] = 96/40 = 12/5 · Var = 12/5 − 9/4 = **3/20**
Median: m³/8 = 1/2 → m³ = 4 → **m = ∛4 ≈ 1.587**. (Mean ≠ median because the graph is not symmetric.)

TRICK

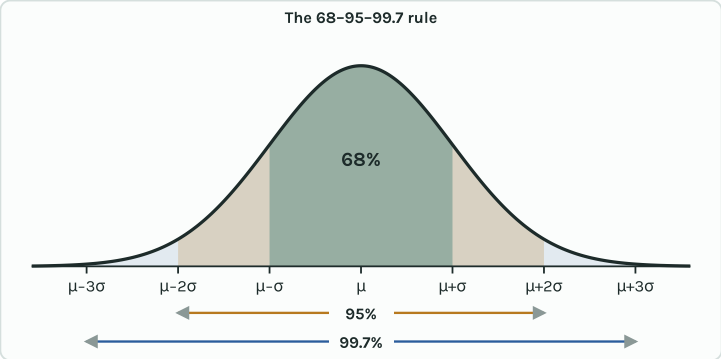
Unknown constant? Set the area to 1

If f(x) = kx on [0, 4], then ∫ kx dx = 8k = 1, so k = 1/8. Then check k makes f(x) ≥ 0.

Try it yourself — Section 6

1. f(x) = x/2 on [0, 2]. Show it is a pdf, then find Pr(X < 1) and E(X).

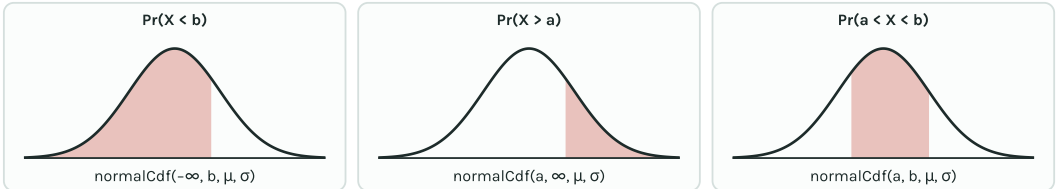
7 The normal distribution X ~ N(μ, σ²)



Fact	What to remember
Shape	bell-shaped, symmetric about μ → mean = median = mode = μ
Notation	`N(50, 9) means σ² = 9, so σ = 3 (not 9!)`
z-score	`z = (x − μ) / σ — how many sds above the mean`
Back to x	`x = μ + zσ`
One tail	Pr(X > μ + σ) ≈ 0.16, Pr(X > μ + 2σ) ≈ 0.025 (halve what is left over)
Exact 95% cut	z = ±1.96 (not ±2)

EXAMPLE — COMPARE WITH Z-SCORES Maths: you got 75, class μ = 65, σ = 5 → z = (75 − 65)/5 = **2**.
English: you got 80, class μ = 70, σ = 8 → z = (80 − 70)/8 = **1.25**.
The Maths mark is more impressive — it is further above its mean in standard deviations.

Calculator commands — always sketch and shade first



You want	Command	Example, X ~ N(70, 4²)
a probability	`normalCdf(lower, upper, μ, σ)`	Pr(X > 75) = normalCdf(75, ∞, 70, 4) = **0.1056**
an x-value from a probability	`invNorm(area to the LEFT, μ, σ)`	Pr(X < k) = 0.9 → k = invNorm(0.9, 70, 4) = **75.13**
upper-tail k	`Pr(X > k) = q → invNorm(1 − q, μ, σ)`	Pr(X > k) = 0.05 → enter **0.95**, not 0.05

TRICK

Use symmetry — three answers from one calculation

65 and 75 are both 5 (= 1.25σ) away from 70. So once you know Pr(X > 75) = 0.1056, you also know Pr(X < 65) = **0.1056** and Pr(65 < X < 75) = 1 − 2(0.1056) = **0.7887**.

EXAMPLE — FIND AN UNKNOWN M σ = 5 and Pr(X < 50) = 0.025. Find μ.

① z = invNorm(0.025) = −1.96 ② 50 = μ + (−1.96)(5) ③ μ = 50 + 9.8 = **59.8**.

Sense check: a small left-tail probability means 50 is **below** μ ✓

TRAP

The calculator wants σ, not σ²

X ~ N(50, 9): type σ = **3**. Write "σ = ..." on the page before you touch the calculator.

TRAP

Standard deviations don't add — variances do

For two independent X_1, X_2 with sd σ, $\text{Var}(X_1 + X_2) = 2\sigma^2$, so sd = σ√2. But 2X (one value doubled) has sd 2σ.

Try it yourself — Section 7

1. X ~ N(100, 15^2). Use the 68–95–99.7 rule to estimate Pr(X > 130).
2. X ~ N(100, 15^2). Pr(X > k) = 0.1. What number do you enter into invNorm?

8 Sample proportions and confidence intervals

LINK TO YEAR 7 — COLLECTING & SUMMARISING DATA

Census vs sample → p vs $\hat{p}$

	Population (census)	Sample
Who	everyone	a group chosen from the population
Proportion	`p — fixed, usually unknown`	`p̂ = x / n — changes from sample to sample`
Year 7 example	shoe sizes of *everyone* in your team	favourite cereal of 500 Australians

Remember the Year 7 skate-park question: surveying 20 people **at** the skate park is a **biased** sample. A bigger n can't fix a biased method — it must be a **random** sample.

The distribution of $\hat{P}$

What	Formula	Example: p = 0.4, n = 100
Mean	`E(P̂) = p`	0.4
Standard deviation	`sd(P̂) = √( p(1 − p) / n )`	√(0.24/100) = **0.0490**
Approx. normal when	`np ≥ 10 and n(1 − p) ≥ 10`	40 ✓ and 60 ✓

TRICK

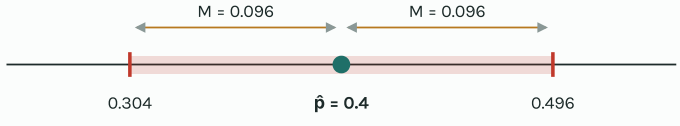
Small n? Turn $\hat{P}$ back into a count

n = 4, p = 0.6: $\Pr(\hat{P} \geq 0.5) = \Pr(X \geq 2)$ where X ~ Bi(4, 0.6) = **0.8208**. (Multiply $\hat{p}$ by n to get the count.) Don't use the normal here — np = 2.4 is far below 10.

Approximate confidence interval for p

`( p̂ − z√(p̂(1 − p̂)/n) , p̂ + z√(p̂(1 − p̂)/n) )` where the margin of error `M = z√(p̂(1 − p̂)/n)`

Confidence level	90%	95%	99%
z	1.645	1.96	2.576
Interval	narrower	→	wider



EXAMPLE $\hat{p} = 0.4$, $n = 100$, 95%. $M = 1.96 \times \sqrt{(0.4 \times 0.6 / 100)} = 1.96 \times 0.0490 =$ **0.096**.
Interval: $0.4 \pm 0.096 =$ **(0.304, 0.496)**.
Backwards: from an interval, $\hat{p}$ = (upper + lower)/2 and M = (upper − lower)/2.

How big a sample?

Rule	Formula	Example
sample size for margin M	$n \geq z^2 \hat{p}(1-\hat{p}) / M^2$	95%, M ≤ 0.03, no estimate of p: $n \geq 1.96^2 / (4 \times 0.03^2) = 1067.1 \rightarrow$ **n = 1068**
no estimate of p (worst case)	use $\hat{p}(1-\hat{p}) = 0.25$	
halve M	multiply n by **4** (to cut M to a third, multiply n by 9). Doubling n only divides M by $\sqrt{2} \approx 1.41$	

TRAP

Round the sample size UP — always

n ≥ 1067.1 means 1068, even though .1 would normally round down. n = 1067 gives M = 0.030001, which is too big.

TRAP

Say what a confidence interval means — correctly

✓ Correct	"We are 95% confident that the interval (0.304, 0.496) contains p." Or: "about 95% of intervals made this way would contain p."
× Wrong	"There is a 95% chance that p is between 0.304 and 0.496." — p is a fixed number; it is the **interval** that changes between samples.

CHECK

Three checks on any interval

① $\hat{p}$ is exactly the middle. ② Both ends are between 0 and 1. ③ On the same data, 99% is wider than 95%, which is wider than 90%.

Try it yourself — Section 8

1. In a random sample of 400 people, 120 like a new app. Find $\hat{p}$ and an approximate 95% confidence interval for p.
2. A poll wants a margin of error of 0.05 at 95% confidence with no estimate of p. What sample size is needed?
3. If the sample size is multiplied by 4, what happens to the margin of error?

Answers — Try it yourself

Section 1

1. $x \leq 4$ and $x \neq -1$ → $(-\infty, -1) \cup (-1, 4]$

2. $(2x + 5)/(x + 1) = 2 + 3/(x + 1)$ → **$x = -1$, $y = 2$**

3. $x = -2$: touches (turning point). $x = 1$: flattens and passes through (stationary point of inflection).

Section 2

1. $2x \in [0, 4\pi]$ → 2 cycles × 2 = **4** solutions

2. Base angle $\pi/3$, cos − in Q2, Q3 → **$x = 2\pi/3, 4\pi/3$**

3. Amplitude 2, period $2\pi/3$, range $[3, 7]$

Section 3

1. ran g = R $\not\subset$ $[0, \infty)$, so no. Restrict to **dom g = $[-2, \infty)$**.

2. x: $1 \div 2 - 1 = -\frac{1}{2}$; y: $-3 + 4 = 1$ → **$(-\frac{1}{2}, 1)$**

3. $\sin(2(x - \pi/6))$: dilation of factor ½ from the y-axis, then translation of $\pi/6$ units in the positive direction of the x-axis.

Section 4

1. $\Pr(A \cup B) = 0.6 + 0.5 - 0.3 =$ **0.8**; neither = 1 − 0.8 = **0.2**

2. $0.6 \times 0.5 = 0.3 = \Pr(A \cap B)$ → **independent**

Section 5

1. $0.2 + k + 0.5 = 1$ → **$k = 0.3$**; $E(X) = 0.2 + 0.6 + 1.5 =$ **2.3**

2. $E(X) = 1$, $\text{Var}(X) = 0.9$, $\Pr(X \geq 1) = 1 - 0.9^{10} \approx$ **0.6513**

3. **$\Pr(3 \leq X \leq 5)$**

Section 6

1. $f \geq 0$ and $\int_0^2 x/2\, dx = [x^2/4] = 1$ ✓; $\Pr(X < 1) =$ **1/4**; $E(X) = \int_0^2 x^2/2\, dx =$ **4/3**

Section 7

1. $130 = \mu + 2\sigma$ → about 0.025 (exact 0.0228)

2. Left area 1 − 0.1 = **0.9**: invNorm(0.9, 100, 15) ≈ 119.2

Section 8

1. $\hat{p} = 0.3$; $M = 1.96\sqrt{(0.3 \times 0.7/400)} \approx 0.0449$ → **(0.255, 0.345)**

2. $n \geq 1.96^2/(4 \times 0.05^2) = 384.2$ → **$n = 385$**

3. It is **halved** ($M \propto 1/\sqrt{n}$)